NRC·CNRC

NRC Publications Archive
Archives des publications du CNRC

Evacuation time and movement in apartment buildings

Proulx, Guylène

This publication could be one of several versions: author's original, accepted manuscript or the publisher's version. / La version de cette publication peut être l'une des suivantes : la version prépublication de l'auteur, la version acceptée du manuscrit ou la version de l'éditeur.
For the publisher's version, please access the DOI link below./ Pour consulter la version de l'éditeur, utilisez le lien DOI ci-dessous.

Publisher's version / Version de l'éditeur:
https://doi.org/10.1016/0379-7112(95)00023-M
Fire Safety Journal, 24, 3, pp. 229-246, 1995

NRC Publications Archive Record / Notice des Archives des publications du CNRC :
https://nrc-publications.canada.ca/eng/view/object/?id=7b1d8203-c479-4e7a-b0ef-d43c7949c5d0
https://publications-cnrc.canada.ca/fra/voir/objet/?id=7b1d8203-c479-4e7a-b0ef-d43c7949c5d0

Access and use of this website and the material on it are subject to the Terms and Conditions set forth at
https://nrc-publications.canada.ca/eng/copyright
READ THESE TERMS AND CONDITIONS CAREFULLY BEFORE USING THIS WEBSITE.

L'accès à ce site Web et l'utilisation de son contenu sont assujettis aux conditions présentées dans le site
https://publications-cnrc.canada.ca/fra/droits
LISEZ CES CONDITIONS ATTENTIVEMENT AVANT D'UTILISER CE SITE WEB.

Questions? Contact the NRC Publications Archive team at PublicationsArchive-ArchivesPublications@nrc-cnrc.gc.ca. If you wish to email the authors directly, please see the first page of the publication for their contact information.

Vous avez des questions? Nous pouvons vous aider. Pour communiquer directement avec un auteur, consultez la première page de la revue dans laquelle son article a été publié afin de trouver ses coordonnées. Si vous n'arrivez pas à les repérer, communiquez avec nous à PublicationsArchive-ArchivesPublications@nrc-cnrc.gc.ca.

National Research Council Canada | Conseil national de recherches Canada

Canada

EVACUATION TIME AND MOVEMENT IN APARTMENT BUILDINGS

Guylène PROULX, Ph.D.
National Fire Laboratory
Institute for Research in Construction
National Research Council of Canada
Ottawa, Ontario, CANADA K1A 0R6

ABSTRACT

An experiment was designed to observe evacuation time and occupant movement in four apartment buildings during a simulated fire emergency. The buildings were 6-7 storeys high containing between 80 and 130 apartments and had an average population of 150 occupants. The buildings chosen were characterized as mixed occupancy housing; that is, they included adults, children, seniors and people with disabilities.

The evacuation drills were recorded on video-cameras located throughout the buildings. The results were analyzed regarding the behaviour of the people, the occupants' time to start to evacuate and their time to reach an outside exit. The impact of the physical organization of the buildings, the evacuation strategies and the occupants' response and movement are discussed.

It was concluded that elderly people and people with disabilities did not impede the evacuation since occupants who were mobility impaired waited in their units to be

rescued. In Buildings 1 and 3, problems arose when the firefighters tried to communicate with occupants remaining in their units because the fire alarm was too loud and most people had moved to their balconies. Buildings 2 and 3 were typical with long delays before evacuation as some mobile occupants did not hear the fire alarm and started their evacuation only when the firefighters knocked on their doors.

INTRODUCTION

Canada and the United States share a common statistic of which they can certainly not be proud; for a few decades now, these two countries have had the highest rate of fires in buildings and the highest fire death rate per capita among all the developed countries in the world [1, 2]. The buildings most affected are residential occupancies [3]. In Canada, the statistics for 1991 showed that 86% of fire deaths occurred in housing, representing 45% of the total number of fires and 45% of the total fire loss [4]. People who die or are injured in residential fires are more likely to be the very young or the very old, according to the U.S. National Fire Protection Association [5, 6]. People with disabilities are also certainly a group at risk, although no statistics currently exist on that population [7].

It appears that the main difference between North American countries and European and Asian developed countries, which have much lower fire death rates, is the emphasis of the latter countries on education and training [8]. In North America, the interaction of public fire safety education, the use of smoke detectors [9], and the

installation of residential sprinklers [10], should eventually reduce the risk to life from fire.

Meanwhile, the profile of housing residents is changing, as society is changing, to include more and more elderly people and people with disabilities in standard residential occupancies. Studying evacuations of such residential buildings with mixed abilities' occupants appears essential to better understand the possible problems firefighters may encounter and to devise plans of action to educate and train occupants, as well as managers, about fire safety in these residential buildings. Furthermore, realistic data on the time and movement of occupants during residential evacuations are desperately needed to be incorporated in computer models, such as FiRECAM [11], to assess the risk to life of occupants from fires.

OCCUPANTS WITH MIXED ABILITIES

In the last decade there has been great concern regarding the accessibility of most people with various disabilities to a large range of buildings. The housing sector was one area where the need for accessibility was acknowledged and where a serious attempt to provide accessibility has been observed. More and more standard residential occupancies have mixed abilities' residents - which assumes a mix of family groups with children of all ages, elderly people and people presenting some level of disability or limitation.

Due to economics and the growing number of people with limitations, it is unrealistic to expect the construction of new specialized buildings where people with disabilities can reside. Furthermore, most people with disabilities and elderly people are determined to stay by themselves and live as long as possible in standard types of housing. It is not surprising then, that more and more standard residential occupancies, contain residents with mixed abilities. The 1991 national Census of Population in Canada shows that 4.2 million Canadians reported some level of disability [12]. This number represents 15.5% of the total population. Among Canadians with disabilities, 93.7% are living in private households compared to 6.3% living in special institutions. Canada is also experiencing an increase in the ratio of people over 65 years of age compared to the total population. The proportion of elderly people rose from 10.7% in 1986 to 11.6% in 1991. Over 91% of elderly Canadians are living in private households [13]. Clearly most disabled people and people over 65 years of age are living in standard housing in Canada. Consequently, it could be expected that approximately 20% of the residents in standard housing are people with some sort of limitation on movement, perception or cognitive capacity.

LIFE SAFETY FOR EVERYONE

The tendency to find occupants with mixed abilities in standard housing is likely to increase as the population ages and people with disabilities are encouraged to live autonomously. Ensuring life safety for mixed abilities' occupants in housing raises new issues. While accessibility is offered, the question of "egressibility" is now becoming a

subject of concern. Egressibility means the possibility of leaving a building or reaching an area of safety in case of an emergency. The egressibility concept does not mean that every occupant should evacuate in the same fashion or through the same route; rather, it intends to promote acceptable life safety for everyone [14].

Before suggesting different ways to ensure life safety for everyone, it is essential to look at the actual situation in residential occupancies. Since there are already numerous residential buildings with mixed abilities' occupants, the assessment of their safety procedures should provide some insights as to what works well and what should be improved. According to local building officials, any recent apartment building ensured an acceptable degree of life safety for all occupants. Full evacuation drills, however, are rarely practiced and have never been studied in such buildings to determine how and how much life safety is provided. To address this issue, an experiment was designed to observe evacuation times, occupants' movements and the impact of mixed abilities occupants in four apartment buildings.

EVACUATION STUDY OF FOUR APARTMENT BUILDINGS

The research involved collecting real data on evacuation times for occupants of apartment buildings. The buildings chosen were located in four different cities to take into account a variety of evacuation strategies used by local fire departments. Each evacuation was organized, supervised and carried out with the participation of the local fire department.

The apartment buildings were characterized as mixed occupancy buildings; the proportion of occupants with limitations in each was approximately 20%. It is important to realize, however, that the goal of the research was not to study the evacuation of disabled persons as such, but to gather data on the time and the manner in which occupants with mixed abilities leave their apartments in standard apartment building settings.

The four buildings studied were concrete and masonry buildings, 6 to 7 storeys high, built between 1982 and 1987. Each of the four buildings was equipped with different fire protection features, had a different physical layout and had a different proportion of adults, children, elderly and occupants with limitations. All were considered, however, to be mid-rise residential buildings according to the building codes. These buildings were required, amongst other things, to have a central fire alarm systems with pull-stations located in the corridors.

Building 1 was different from the other three as it was equipped with a two-stage fire alarm system. This meant that, upon activation of a manual pull-station or a fire detector, an alert signal (intermittent rings) would sound throughout the building. This signal would change into a full fire alarm (continuous ringing) after 5 minutes. Every apartment had a fire alarm bell inside the apartment in the bedroom. This building was also equipped with a public address system that allowed the firefighters, or people in

authority, to convey messages to the occupants. The P.A. system was linked to speakers located in the corridors and staircases but not in the apartments.

The other three buildings had single-stage central fire alarm systems with alarm-bells located in corridors and staircases. The alarm-bells in Building 2 were recessed into the corridor walls. In Building 3, the alarm-bells were installed on the corridor walls. Building 4 had a complex architectural design, with alarm-bells located only on the 4th and 6th floors, and in the staircases. Figure 1 is a floor plan for Building 3, typical of these four buildings.

Figure 1: Floor plan of Building 3

METHODOLOGY

Before each evacuation drill, the residents with some limitations were consulted to assess their knowledge of the building evacuation procedures and their ability to participate unaided in the drill. The initial goal of these meetings was to inform the occupants of the evacuation study procedures and to prepare for the drill, determining potential problems that could occur during the exercise. During the drill, the occupants were expected to follow the fire evacuation procedures planned for their building. It was found that no specific instructions for non-mobile occupants were planned, so these occupants were instructed to stay in their apartments and wait to be rescued by the

firefighters. They received further explanations as to how firefighters would evacuate them horizontally, moving them to a safe area on the same floor [15].

A week before the drill, all occupants were informed, in writing, of the upcoming exercise through a memo distributed to each apartment. The message did not specify the exact day and time of the evacuation drill, however, it highlighted the importance of the exercise and the participation of researchers and the local fire department. It specified that the drill would be recorded by video cameras positioned in hallways and stairwells.

Each drill took place on a weekday between 6:45 pm and 7:30 pm, in order to have the largest possible number of occupants at home, yet before bedtime for most people. The evacuation drills were carried out during the summer and early fall of 1993. For each evacuation, the weather was sunny and warm.

During each drill, data were collected through 12 video cameras. The cameras recorded such events as the time to respond to the alarm, the location, time and frequency of movements and the interaction between occupants. From these recordings, various statistical calculations were performed [16]. During the evacuation drill, the research team stayed outside the building so as not to interfere with either the building occupants or the firefighters.

The local fire department participated in the drill according to a fire scenario planned with them. For all buildings, the fire was located on the 4th floor. After

receiving the call, firefighters arrived in full gear with their trucks and performed their duties according to their normal procedures for such a building. People with disabilities were usually moved horizontally to a safe area, while mobile people were asked to evacuate the building to the outside using stairs. When the firefighters judged that everyone had reached a safe location, the Captain gave the "all clear" signal.

After the evacuation drill, a questionnaire was distributed to each apartment. Occupants were asked if they had heard the fire alarm (and the P.A. messages for Building 1), to list the chronology of their actions after hearing the fire alarm and to describe their evacuation movements.

RESULTS[1]

Prior to the evacuation drills, interviews were carried out in Buildings 1, 2 and 4 with most of the elderly and any occupants presenting some limitations. In Building 3, the building management refused meetings with residents for fear of disturbing or frightening them. The meetings which were meant to gather information on the occupants' capacity to participate in the drill, rapidly became information sessions to reassure occupants and explain some basic fire safety notions. Evacuation drills had never been performed in these buildings, except for one exercise in Building 1 a few years before. The evacuation procedures for the four buildings did not provide any

[1] This paper presents a summary of the results of the study. A full description of the study methodology, results and analyses for individual buildings and comparisons among buildings are presented in an NRCC Internal Report [17].

specific directives for people with mobility limitations. During the meetings, people generally mentioned they had not received any instruction but they thought they were supposed to go on their balcony and wait to be rescued. Many elderly residents and people with disabilities wondered if the firefighters would carry them down the stairs or down a ladder during the drill. In Buildings 1 and 2, each person met said that he/she had heard the fire alarm previously and could recognize it immediately. Some occupants had already been through the evacuation procedure due to false alarms (or two small fires in Building 1). The majority of occupants in Building 4 had never previously heard the alarm in that building.

THE TIME TO START

The results for the time to start the evacuation for the four buildings are presented in Table 1. This time represents the elapsed time between the fire alarm sounding and the moment the person left his/her apartment.

Table 1: Time to Start the evacuation

As presented in Table 1, the time to start to evacuate varied considerably among the buildings. The statistical analysis of variance (ANOVA) shows significant differences among the four buildings. There is, however, no significant difference between Buildings 1 and 4 where occupants started rapidly and Buildings 2 and 3 where occupants took more time to start.

An analysis of the questionnaire shows that 100% of Building 1 occupants stated that they heard the fire alarm since the alarm-bells were located inside each apartment. In Building 4, 17% stated that they did not hear the alarm. These occupants were located on the 1st and 2nd floors, the closest alarm-bells were located in the staircases or on the 4th floor. The occupants who did not hear the alarm started their evacuation later when they heard or saw the firefighters arriving. For Buildings 1 and 4, every evacuee had started to evacuate within 15 minutes of the alarm sounding.

ANOVA's show that Buildings 2 and 3 present no significant difference for the time to start. The questionnaire shows that 25% of the occupants of Building 2, and 23% in Building 3 did not hear the fire alarm from inside their apartments. These occupants had to be alerted by neighbours or firefighters knocking at their doors. Some people took as long as 25 minutes before starting to evacuate.

The time delay to start evacuating for 64% of Building 1 occupants and 52% of Building 4 occupants was 2:30. This time delay appears to be the time needed for occupants to perform pre-evacuation actions upon hearing the fire alarm. In Buildings 2 and 3, where the alarm-bells were not heard by all occupants, the time delay to start evacuating extended, for some occupants, to the time at which the firefighters arrived, 5 to 8 minutes after the alarm activation. Table 2 shows the average delay time to start the evacuation.

Table 2: Average time to start

These results show that some time was spent by occupants in all buildings in pre-evacuation actions. The actions favoured, according to the questionnaire, are presented in Table 3.

Table 3: Pre-evacuation action

It is interesting to note that only occupants of Buildings 2 and 3 mentioned the action "have a look in corridor." This suggests that these occupants had to move to the corridor to confirm the fact that the alarm had activated. Following that, they undertook other actions before starting to evacuate.

The public address messages in Building 1 were understood by only 15% of the people. As mentioned in different studies [18, 19], it is essential to ensure that occupants can clearly hear and understand messages from a P.A. system; otherwise, it is useless. Worse, it can lead to misunderstanding instructions.

Overall, from the ANOVA calculations, the age and gender of the occupants made no significant difference to the time to start the evacuation. The time to start cannot be calculated for the occupants with mobility limitations since they were instructed to stay in their apartments and wait to be rescued. It is interesting to note though, that people with hearing impairments had no special devices to inform them of

the fire alarm activation. These people did not evacuate unless they were informed directly by another person.

THE TIME TO EVACUATE

The time to evacuate is the time occupants spend between the sounding of the fire alarm to the time at which they reach an area of safety. For most of the occupants, it is the time at which they cross an outside exit. A few occupants took refuge on their balconies but they are not included in the analysis for the time to evacuate. The time to evacuate is closely linked to the time to start; occupants who started late had a longer evacuation time.

Table 4 presents the average time to evacuate for the four buildings. Among all the buildings, the ANOVA tests show significant differences between the mean times to evacuate. The mean time to evacuate was not significantly different between Buildings 2 and 3. The average time to exit these buildings is respectively 9:36 and 10:57, which is very long, but many occupants in these two buildings started very late. Buildings 1 and 4 are different for the time to evacuate since Building 1 had a significantly lower exit time with an average time of 3:05 compared to 4:38 for Building 4.

Table 4: Average Time to Evacuate

An interesting statistic, presented in Table 4, is the time to move which represents the time it took for a person to reach an outside exit from the moment the person left the apartment, regardless of the distance travelled. The ANOVA's conducted show that there is no significant difference among the four buildings. The average time to travel from an apartment to an exit was just over 1 minute for all occupants of the four buildings.

Overall, the results of the ANOVAs show that gender had very little influence on the outcome of the evacuation times for all buildings. There were differences between the times to exit for males and females; the males generally took more time. None of these differences was significant, however.

Looking at all four buildings, age was not a major factor in determining the time to evacuate. In some cases, as can be expected, the younger people moved faster than the older individuals, but this result did not always hold true. Basically, there were very few significant time differences among the age groups. Interestingly, in two of the buildings, seniors moved faster in starting their evacuation but moved slower in the corridors and the stairs. Eventually, they reached safety in approximately the same mean time as people from the other age groups. Occupants under the age of twenty could not be used in the statistical analyses because there were too few in these age groups.

The speed at which a person can travel is determined, in part, by the extent of his or her physical ability. The duration of an evacuation, then, can vary depending on the

abilities of the building occupants. All four buildings in this study had occupants with mixed-abilities. Because of this, there were many occupants in the buildings with some disability or limitation that could hinder them or others while walking, especially down stairs. One of the purposes of this research was to examine evacuation times for those individuals with some limitation. This was an attempt to determine whether significant differences existed in evacuation times between those with limitations and those without.

Through an examination of the video tapes, the occupants were placed in either a "limitation" or "no limitation" group based on their general behaviour during the evacuation. The characteristics of the people in the "limitation" group were: walked with a cane, slow walker, visually impaired and persons with multiple sclerosis. Two other categories were included in the group with limitations; those carrying children and occupants over 65 years old who had no outward signs of impairment but who were likely to have some sort of limitation due to their age. It should be noted that wheelchair users and other mobility impaired people were not included in the evacuation time analysis because most of them remained in their apartments until firefighters moved them to an area of safety.

ANOVAs were conducted for all four buildings, comparing the times of those with limitations to those without limitations. The results for Buildings 1 and 4 cannot, however, be used because they had less then 5 people with limitations who actually evacuated the building - which is very difficult to compare to a much larger group of people without limitations. Since Buildings 2 and 3 had no significant differences in

mean times, occupants with limitations were combined to make a larger group of subjects. Occupants without limitations in Buildings 2 and 3 were put in another group. ANOVAs were conducted comparing these two groups with the results shown in Table 5.

Table 5: Time Comparisons for Occupants with Limitation vs. No Limitation from Buildings 2 and 3

As can be seen in Table 5, the occupants with limitations were slower than those without limitations for the average time to start and to evacuate. These differences, however, are not significant. This result was expected for the time to start, since, in a small, familiar place such as an apartment, those with limitations are usually able to move quite readily. A significant difference does exist between the two groups for the time to move. Once people left their apartments, the stairs seem to pose a problem for those with limitations and, as a result, they moved significantly slower than the others.

SPEED OF OCCUPANTS ON STAIRS

The use of video cameras provided invaluable information on the movement of occupants on stairs. The mean time to descend one flight of stairs (from floor to floor) was determined for the occupants of all four buildings. Not all occupants could be used in these calculations since some exited from the main floor and, therefore, did not use the stairs. Others were omitted from this analysis because they stopped repeatedly in the staircase to chat with neighbours. The average speed of occupants measured along the line of the stairs is presented in Table 6.

Table 6: Speed of occupants on stairs

It is important to note that the stairs were never crowded during the evacuation. The average speed shown for Building 2 (0.44 m/s) is based on an estimated measurement of the stair dimensions. It was impossible to obtain a speed for Building 4 because there were three stairwells of complex design, which varied in stair width, number of steps between landings, shape, size, and number of landings. Occupants in three of the four buildings studied had a comparable mean speed on the stairs; they travelled between .41 m/s and .47 m/s on non-crowded stairs.

SOME OTHER RESULTS

Among the four buildings, there were five occupants who evacuated down the stairs while carrying children. Their speed on stairs varied from .38 m/s to .57 m/s which is comparable to the general average speed. It was observed, however, that these people had to take extra precautions while going moving the stairs. They were usually carrying the child on one hip while holding the handrail with their free hand. The absence of crowding permitted the use of lateral space, as needed, which probably contributed to their fast speed of movement.

Small children (aged 2-5) were among the slowest groups during the evacuation. Their speed on the stairs was .33 m/s which is similar to the speed of occupants over 65 years old, (.35 m/s). Children aged 2-5 were not carried by adults. They descended by themselves, one step at a time, holding the handrail with one hand at a height over their

shoulders, an adult holding their free hand. They occupied the full width of the staircase. The groups with small children would have considerably slowed down the evacuation of other descending occupants if there had been a crowd on the stairs.

Through the analysis of the videotapes, it became apparent that people travelled in groups during the evacuations. These groups were mainly pairs or groups of three. Families with children would typically evacuate in a close group with an adult carrying the smallest child. The family groups split up slightly as the children increased in age. Children, six and older, travelled with the family but were usually four or five steps ahead racing down the stairs.

Although some seniors evacuated individually, they usually travelled in groups of two or three. Often they would exit their apartments and gather to discuss the fire drill and then proceed to leave the building in a group.

Among the four buildings, 62% of the occupants evacuated in groups. These group formations likely delayed the speed of movement of the group because members tended to assume the speed of the slowest person. In most cases, the slowest person was a young child or an elderly person. Older people also tended to stop to converse rather than maintain the same speed during the evacuation.

In all four buildings studied, the occupants had a choice of at least two staircases when evacuating. The different sets of stairs were not used equally as means of egress.

It is generally believed that people will use the closest stairs to evacuate, however, more progressive concepts such as "the affiliation model" predict that people will move toward areas with which they are most familiar [20]. Areas around the elevators and stairs leading to main exits are seen as familiar. People may use stairs that are less convenient in proximity to their apartment simply because they spontaneously go to areas with which they are familiar. Table 7 shows the percentage of occupants using each staircase in the different buildings.

Table 7: Use of staircase

The occupants of Building 1 had a choice of only two staircases located at extreme ends of the rectangular building. As the table shows, the majority of the people used the right staircase because the firefighters who arrived very early during the evacuation directed the occupants toward that staircase. In Building 2, the distribution of the occupants during the evacuation was more uniform for the three staircases. The left staircase was the least used since it was located at an extreme end of the building and led to the side of the building. The centre staircase and the right staircase were mainly used since the centre staircase led to the front, while the left staircase was a familiar exit leading to a garden and play area.

In Buildings 3 and 4, the centre staircase was used by 60% and 75% of the evacuees, respectively. For both buildings, the centre staircase was located in the middle of the rectangular structure near the elevators leading to the front entrance. This was the

familiar direction of movement to leave the building for occupants, which confirms the affiliation model in these evacuations.

CONCLUSIONS

The results of this study show that the inability of people to hear the fire alarm in their apartments is a major factor in the time to start an evacuation. In two of the buildings studied, 25% of the occupants could not hear the fire alarm in their apartments. Consequently, those occupants started their evacuation only when the arriving firefighters knocked on their doors, providing a mean time to start the evacuation of 9:02 for these two buildings. It is obvious that the location of the alarm-bells has a major impact on the audibility of the fire alarm [21]. Locating the alarm-bells in the corridors and staircases does not ensure that occupants will hear the alarm from inside their apartments. Such alarm-bell locations also seem counter-productive, since, once in the corridor or the staircase, the sound is so loud that the exchange of verbal information becomes impossible and occupants cannot discuss a plan of action. Furthermore, the mobility impaired people, who are moved to a safe such as staircase landing or the elevator lobby, have to suffer the deafening sound of the alarm. Alarm-bells with a lower sound, in every apartment, appear more efficient to alert occupants to a possible fire.

In the two buildings where the alarm was audible to most occupants, the mean time to start was 2:49. This time is probably a good representation of the time used by occupants to prepare to evacuate their apartments. Questionnaires showed that this time

delay was used to get dressed, gather valuables, find children, or find pets. The time delay to start evacuating is important and should be taken into account in fire safety assessment models to give a more realistic representation of human factors during evacuation and to better assess life safety in buildings.

In buildings where a longer time to start was observed, there was, consequently, a longer time to totally evacuate the building. Due to the very low occupant density in this type of building, there was no queuing to enter the staircase and no crowds in the stairs. Therefore, the time to evacuate was dependent only on the time to start and the distance to travel.

Overall, the statistical analyses showed that gender had little influence on the outcome of the evacuation for all buildings. There were differences between the speeds of males and females, but none of these differences was significant. Age was not a dominant factor either. In some cases, the younger people moved faster than the older ones, but this result did not always hold true. Basically, there were very few significant time differences among the age groups. Interestingly, in two of the buildings, seniors moved faster in starting their evacuation but moved slower in the corridors and the stairs. Eventually, they reached safety at approximately the same mean time as people in the other age groups.

Altogether, whether the occupants had some sort of limitation or not did not have a great impact on the evacuation timing. People with limitations, who evacuated, were

generally slower than others but this difference was only significant during the actual moving time outside the apartments. The overall evacuation time, however, was not significantly affected by the presence of people with limitations because most occupants with mobility limitations stayed in their apartments waiting to be rescued by arriving firefighters.

An unexpected problem arose with mobility impaired people who went out to their balconies and waited to be rescued. On one hand, this procedure worked very well since the people with mobility impairments did not try to leave by themselves and did not impede the evacuation of others. On the other hand, however, people who moved to their balconies could not hear the firefighters knocking at their doors to rescue them. Some alternative way to communicate with people who stay in their apartments has to be developed to ensure that instructions can be heard and understood by people who need assistance.

Occupants of three of the four buildings showed a comparable mean speed on the stairs. Among all occupants, they travelled between .41 m/s and .47 m/s on non-crowded stairs. Those over 65 travelled significantly slower at .35 m/s; children between the ages of 2 and 5 travelled at a mean speed of .34 m/s.

Most occupants tended to evacuate in groups at the speed of the slowest member of the group. While evacuating, occupants tended to use either the most central staircase,

located in their familiar path of travel, or a stairwell that led to a familiar area, such as the main entrance.

The results showed that evacuation time does not vary considerably among evacuees. In fact, the majority took about 1 minute to travel the distance between their units and an outside exit, regardless of the location of their starting points. This indicates that, in a low-rise apartment building the distance and time to reach an outside exit are of secondary concern. In fact, what is very important is the starting time to evacuate. From this study, it appears essential to develop strategies to shorten the time to start the evacuation as a means to reduce the total evacuation time of all occupants. More information and training would be useful to make people understand that immediate response to an alarm is important for their own safety and the safety of others. Education should be provided on what actions are expected from occupants when the fire alarm sounds, specific procedures have to be planned for mobility impaired occupants. The number of false alarms has to be reduced to rebuild the faith in occupants about the meaning of an alarm.

In one of the buildings studied, occupants had not received any information prior to the drill. In that building, many elderly people and mobility impaired residents were unsettled by the fire alarm, not knowing what to do. A lady in a wheelchair entered the main staircase blocking the way to descending occupants, threatening that she was going to try to go down the stairs with her chair. Residents took her back to the corridor and stayed with her until the firefighters arrived. The uncertainty and anxiety observed

during that evacuation demonstrated the importance of training and information on evacuation and fire safety if appropriate evacuation behaviour is expected during an emergency.

The results of this study will be used in the development and validation of computer models, such as FiRECAM being developed at the National Research Council of Canada. The use of ecological data will provide models with strong predictive capabilities to assess life safety of occupants in apartment buildings.

ACKNOWLEDGEMENTS

The author wishes to thank the Ottawa, Montreal, East York and Vancouver Fire Departments for their willing participation in this study. Thanks also to John W. MacLaurin and John C. Latour for expertly managing the technical side of this project, and to Curt McQueen for the data analysis. This research was jointly funded by the Canada Mortgage and Housing Corporation and the National Research Council of Canada.

REFERENCES

1. P. S. Schaenman, International Concepts in Fire Protection, Ideas from Europe that could Improve U.S. Fire Safety, TriData Corporation, Arlington, VA, U.S.A., 1982, 93p.
2. J.R. Hall, Fire in the U.S.A. and Canada, International Fire Comparison Report #2, National Fire Protection Association, Quincy, MA, U.S.A. March 1994, 11 p.

3. K. J. Tremblay, Catastrophic Fire Deaths: The Numbers are Back Up, NFPA Journal, 88(5), National Fire Protection Association, Quincy, MA, U.S.A., Sept./Oct. 1994, pp. 88-109.
4. Association of Canadian Fire Marshals and Fire Commissioners, Annual Report 1991, Fire Losses in Canada, Minister of Labour, Government of Canada, 1993, 41 p.
5. A. L. Miller, Patterns of Fire Casualties in Home Fires by Age and Sex, 1986-1990, National Fire Protection Association, Quincy, MA, U.S.A., June 1993.
6. J. S. Petraglia, Fire and the Aging of America, NFPA Journal, 85(2), National Fire Protection Association, Quincy, MA, U.S.A., March/April 1991, pp. 36-46.
7. C. J. Conley, & R. F. Fahy, Who Dies in Fires in the United States?, NFPA Journal, 88(3), National Fire Protection Association, Quincy, MA, U.S.A, May-June 1994, pp. 99-106.
8. P. S. Schaenman, International Concepts in Fire Protection, New Ideas from Europe 3, TriData Corporation, Arlington, VA, U.S.A, 1993, 68p.
9. J. R. Hall, The U.S. Experience with Smoke Detectors: Who has Them? How Well Do They Work? When Don't They Work?, NFPA Journal, 88(5), National Fire Protection Association, Quincy, MA, U.S.A., Sept./Oct. 1994, pp. 36-46.
10. J. R. Hall, "The U.S. Experience with Sprinklers: Who has Them? How Well Do They Work?", NFPA Journal, 87(6), National Fire Protection Association, Quincy, MA, U.S.A., Nov.-Dec. 1993, pp. 44-55.
11. D. Yung, & G.V. Hadjisophocleous, The use of the NRCC Risk-Cost Assessment Model to apply for code changes for 3-storey apartment buildings in Australia, Proceedings of the Symposium on Computer Applications in Fire Protection Engineering, Society of Fire Protection Engineers, Boston, MA, 1993, pp. 57-62.
12. Statistics Canada, 1991 Census Handbook, Catalogue 92-305E, Statistics Canada Ottawa, 1992.
13. Statistics Canada, Dwellings and Households, Catalogue 93-311, Statistics Canada Ottawa,1992.
14. E. Juillet, & J. Pauls, Recent Social and Technical Developments Influencing the Life Safety of People With Disabilities: The North American Scene, Proceedings of

the IAPS 12th International Conference, Marmaras Chalcidiki, Greece, 11-14 July, 1992.

15. H. J. Klote, H. E. Nelson, S. Deal, & B. M. Levin, Staging Areas for Persons with Mobility Limitations, NISTIR 4770, U.S. Department of Commerce, National Institute of Standards and Technology and General Services Administration, Gaithersburg, MD, 1992.
16. G. Proulx, & C. McQueen, Evacuation Timing in Apartment Buildings, Internal Report No. 660, Institute for Research in Construction, National Research Council of Canada, Ottawa, June 1994, 20 p.
17. G. Proulx, J. C. Latour, & J. W. MacLaurin, Housing Evacuation of Mixed Abilities Occupants, Internal Report No. 661, Institute for Research in Construction, , National Research Council of Canada, Ottawa, July 1994, 50 p.
18. G. Proulx, J. D. Sime, To Prevent 'Panic' in an Underground Emergency, Why Not Tell People the Truth?, Proceedings of the Third International Symposium on Fire Safety Sciences, Elsevier Applied Science, London, U.K. 1991, pp. 843-852.
19. J. P. Keating, E. F. Loftus, & M. Manber, Emergency Evacuations during Fires: Psychological Considerations, R. F. Kidd & M. J. Saks (eds.), Advances in Applied Social Psychology 2, Lawrence Erlbaum Associates Pub., Hillsdale, NJ, 1983, pp. 83-99.
20. J. D. Sime, Movement Toward the Familiar Person and Place Affiliation in a Fire Entrapment Setting, Environment and Behavior, Sage Publications, Newbury Park CA, 17(6), 1985, pp. 697-724.
21. M. A. Sultan, & R. E. Halliwell, Optimum Location for Fire Alarms in Apartment Buildings, Fire Technology, 26(4), 1990, pp. 342-356.

Figure 1: Floor plan of Building 3

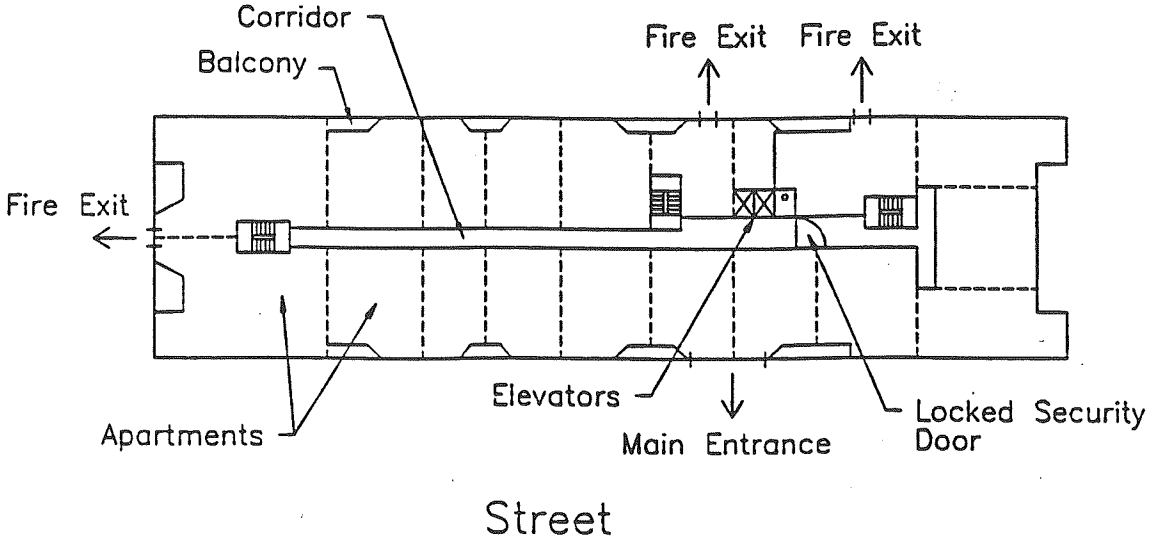


Table 1: Time to Start for the first 5 minutes

Time Intervals	Building 1		Building 2		Building 3		Building 4	
	Frequency	%	Frequency	%	Frequency	%	Frequency	%
0:00-1:00	11	26	5	9	8	10	17	21
1:01-2:00	14	33	10	18	12	16	13	16
2:01-3:00	7	17	8	15	6	8	26	33
3:01-4:00	4	10	3	5	5	6	9	11
4:01-5:00	3	7	3	5	3	4	4	5
Total		93%		53%		44%		86%

Table 2: Average Time to Start

Building	Time to Start min:sec
1	2:30
2	8:22
3	9:42
4	3:08

Table 3: Pre-evacuation action

Building	%	Action
1	19	"find pet"
	17	"gather valuables"
	15	"get dressed"
2	34	"have a look in corridor"
	21	"move to balcony"
3	23	"gather valuables"
	20	"have a look in corridor"
4	30	"get dressed"
	13	"find children"

Table 4: Average Time to Evacuate

Building	Time to Evacuate min:sec	Time to Move min:sec
1	3:05	1:05
2	9:36	1:17
3	10:57	1:15
4	4:38	1:07

Table 5: Time comparisons for occupants with Limitation vs. No Limitation of Buildings 2 and 3

Time Level	Limitation min:sec	No Limitation min:sec	p-value
Time to Start (mean)	10:34	8:49	p=.3314
Time to Evacuate (mean)	12:06	10:05	p=.2872
Time to Move (mean)	1:37	1:12	*p=.0167

*denotes significant difference in mean times.

Table 6: Speed of occupants on stairs

Building	Mean Time to Descend One Floor sec.	Speed metre/sec.
1	16.1s	0.47 m/s
2	16.4s	0.44 m/s
3	16.6s	0.41 m/s
4	9.6s	*---

* unable to obtain stair distance between floor

Table 7: Use of staircase

Building	Left stair	Centre	Right stair
1	32%	N/A	68%
2	20%	39%	41%
3	25%	75%	0%
4	20%	60%	20%

Table 7: Use of staircase

Building	Left stair	Centre	Right stair
1	32%	N/A	68%
2	20%	39%	41%
3	25%	75%	0%
4	20%	60%	20%